

ActInf Textbook Group ~ Cohort 5 ~ Session 11 (Chapter 5, Part 2) ~ 11/28/2023

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 11 (Chapter 5, Part 2) ~ 11/28/2023 | 54:56

all right welcome back everyone um it's

our

last discussion for this interval of

activity with cohort

5 we'll discuss chapter 5 but also

anything in the first half of the book

just anything of what we've gotten up to

so

far

then next week is our final discussion

for the year we'll look at project ideas

so here's where people have um written

different different ideas that they they

might want to work

on and also we'll review future textbook

groups everyone's welcome to fill out

this Anonymous form and then also it's

always cool to

um see how

many ideas people have had about the

future textbook groups and what to tell

other

participants and also know that in 2024

when we return it's possible for people

who want to explore having like a role

in the textbook group we can explore

that but that'll be for next week so for

today

um we'll go to chapter 5 or again any

section of the book so does anyone want

to just begin with

any thought on five or

anything that they've gotten up

to in the first 100 Pages Susan go for

it so I'd like um a little more

elaboration on the high road and the low
road I better understanding of how to
differentiate those
two
yeah does anyone want to give a thought
on
that I um I still like one thing I said
when we discussed this before um and it
might be worth
dialogue um the low road to me is all
about what is the nature of
inference um so how do
we make sense of the world when the
world is when we don't
have I guess the base
axioms um and then the high road is all
about the nature of identity what does
it mean to be an entity in a world and
what's what are the first principles
that constrain the capacity to be an
embedded entity so it's really the the
nature of the first question you ask and
then um you have a bunch of follow on
propositions that you get from that
first question and both roads lead you
to the same place which is active
inference I like that so this so that's
the um kind of what differentiates I
guess um yeah perception and action so
um so when you say the nature of
identity
um um is identity and you might be
saying it on a couple of different
levels there um you 'You' got the
identity of who am I but
also
um you can expand that into groups and
how they Define their boundaries am I

tracking that the same way that you're thinking

Andrew I would say so and I you can also take an

intrinsic view is just like when you get a sensory map how can you delineate it into separate things um and there's a process of identity going on there that allows you to label your sent field as a collection of separates so identity has like anytime that you can tokenize your senty field into something you've got a a an identity relationship going on can I ask a um what might sound like a

um yeah um so where would you put degrees of freedom you where would you put degrees anytime you have multiple identities

interrelate then you're going to have have a property of degrees of freedom of how they relate so I I guess you could pull it from either direction so it was almost in affordance yeah yeah

okay and Daniel I'm just riffing here so yeah go for it okay yeah that's that's I

have okay all all thanks for that I'll I'll give a little bit of a different um angle um the low road how you'd know you're on it is anytime some specific implementation is being described so that's like the actual constructive materialized basis of some

inferential system it uses base theorem
but JF clout here uses symbolic logic
there's potentially other non- basing
approaches people could
try coming from the high road is the why
you know you're on it because it's
implementation Independence it's
agnostic or factorized with respect to
how you build it so these are talking
about properties of different kinds of
systems like system science or
complexity
science just general system
properties that don't specifically say
how it's built like I want the fast car
that gets me from A to B I want a car
made of
aluminum but maybe the car made of
aluminum is just a museum car and maybe
the car that gets you from made of B is
like made of paper um that's one answer
how on the low road with the material
substrate of the synthetic intelligence
why on the high road um a little
bit previous of a framework that was
used before in the um like late 2010s
when friston started using the high road
low road um
distinction there was uh an appeal
actually to the more classical
partitioning of cause with Aristotle oh
funny typo with Aristotle's for cause
material efficient formal and final so
material cause is like the actual
material like there's a house there
because there's planks of wood the
efficient cause the construction of the
house the formal cause that which makes

it a house and then the Tios or the
final cause and this was used in the
paper answering schinger question free
energy formulation by ramstead at all
2018 and so here they explicitly took
that kind of this is earlier on in their
multiscale
thinking and they
explicitly applied this
to tinbergens for Y which were like the
biological adaptation of aristotles for
wise
so it it's especially when it's
presented um this way in the
textbook it's not for my understanding
intended to be like necessary or
sufficient or say that there's some one
in only relationship between surprise
minimization must be exactly right here
between these two terms it's more just
evocative principles and
and hows that are actually used
but if one understands all the house if
you understand all the building
materials and all the
properties
then
like the how and the why of active
inference on this one road are fully
defined so
so that leads me
to
um
so the in the context of I in fact if I
I kind of wrote down the context of
intent is actually the high road and the
low road um would
be you how I'm going to follow that the

the scope of
actually the um and I'm I'm toying or
playing with or trying to understand
discreet versus
continuous and and I'm I'm I can see
where they are um they are somewhat
embedded in each other so it can't be
delineated exactly but uh but in terms
of the generative
model modeling process is that what is
that what this is pointing
toward this
is like a Prelude to choosing whether
you are going to use a discrete or a
continuous
model this is discreet or continuous AG
agnostic like you could you could go up
and down yeah you could go up and down
and you could have a fully discrete
system or model and you could also have
continuum time continuous steps back yeah
or you could have the hybrid model
okay but it's an interesting question
like is the continuous or discreet just
a little difference or is it a big
difference what this
in so that's where I'm I'm trying to
kind of put the scope of
it
um and and I may be ever complicating it
but um but no thank you for that that's
helpful yes thank you
yeah especially as we come back to
actual systems of interest like in
chapter
five it's just interesting like okay
this is kind of this is the road map in
theory but where's any real system on

this

like do we drive to active

then take take a turn and head over to

the brain or is this our road map that

we then transpose onto the

brain I think there's a lot of space

with like H how these

more evocative

schematics end up being used in a real

setting can I take a I think it's more

of a pedagogical skill tree rather than

an implementation road map

I agree with that I agree with

that um though not explicitly linearly

pedagogical but yes it's it's um like a

mind

map and it's a it's the map and not the

terrain but at the same time it's not

paved it's not bab it's a single Lane

dirt road Pony Express from f p down to

B yeah and so when you when we use the

term policy and I'm I may be conflating

that with strategy and you know so and

I'm I may be reifying the low road with

strategy you say anything about

that so policy is strategy contained in

policy

strategy let's even see if we

use obviously it's not like a word has

to be in this list to to be useful

strategy can refer to a lot of different

things policy most narrowly refers to

the sequence of actions for a given time

Horizon sequence of affordances that are

planned to be taken over a given time

Horizon so that would be down here with

planning as inference so it's like

the two fundamental features or or

attributes that the thing has to have
inbound dependencies which we interpret
as perception outbound dependencies
which we interpret as action now not all
things plan like a pendulum is active
but you don't need to appeal to
planning whereas in the case of things
that do appear to to either
make sentient or Intelligent Decisions
in the moment even if they're just one
time step judgments or specifically
engage with what appears to be planning
like explicit considerations of multiple
time steps and evaluation against
Alternatives and abductive reasoning and
all of

this then how is active inference going
to how hash low road How is active
inference going to approach this
question of planning and action
selection as an inference problem in
fact part of a unified inference problem
with

perception so action selection and
planning and policy selection is down
here in the
basian

branch and then the way that we would
kind of touch to policy or say say what
what is the strategy by which the
actions are selected coming from the
high road without talking about how
we're going to do that
inference it would be like well if
something is going to be self-organizing
itself to persistently minimize its
surprise amidst environmental
regularities

irregularities maintaining its Markov
blanket
and self- evidencing its own existence
on its own body and Beyond through Niche
construction
then it might have to do some kind of
planning where it might have to is like
saying like you know it'd be nice if
somebody made this
dish but it's high road because you're
not saying how that's going to
happen
um any anyone who
hasn't raised anything like want to type
or ask a
question
here's a nice part of five that that
they're discussing it in the
neuromodulation
case however
this applies outside the brain like why
why would we even make maps why would we
associate features of our
Maps generative models with
anything that is actually there in the
territory like why do
it it lets us explain observe
biology form hypotheses and develop
non-invasive
methods also
not um not only or exhaustive like
another kind of common refrain would be
like explain predict design control or
appreciate influence
control developing explanatory
accounts that is about kind
of getting our desk organized organizing
the data we already have what we already

know developing a more coherent or more
concise
account forming
hypotheses
and and developing methods are moving
more into the forward-looking
space you could do that without making
an account
first but usually catching up with the
past then
helps develop the future going
components of the scientific
aspect like a given account of the brain
here's what the brain does and then
here's the methods that people bring to
bear

Susan I really like

how you

Associated um the biology with
self-organization um observed biology so
that brings up for me the observed
biology um is it we're not limiting this
to what's what's
observable or are

we for the mapping for the generative
process mapping I mean I you it was
helpful for me quite frankly to listen
to the session you just posted on
chapter six
yesterday because it kind of brought
home what is this
for and so that generated a lot of new
questions so you know my question when
you say obs OB served
biology

um and you forming hypothesis is a
little bit uh more
tangible but observed biology that that

term yeah it's like okay that's where
the self-organizing
is yeah is the structure it's a funny
way to say it like um that it's observed
I wonder if it's almost like a little
um like let let's think about the
fundamental challenge of the a
matrix to take
observations and map them to Hidden
States so now thinking about the
scientific Endeavor or scientist or just
sense maker in terms of active inference
there's observations and and then
there's the
unobserved hidden States there's like
the estimates of what those
observations reflect thermometers in the
room measurements and then the mapping
to the real temperature so talking about
let us explain observe
biology observed connects us to
empirical
observations like well the person who
was coughing the PCR test had this
result the person who wasn't coughing
the PCR test had that result that's an
observation but then the explanatory
account obviously includes things that
aren't simply
observed otherwise the quote data would
speak for themself but they never do
that's why there's like science and law
and all of
this so This points both ways because
it's saying we are trying to develop an
account of observations now those might
be super super sparse observations we
have one video of a Jaguar one time

doing this
move so it's not that observation means
we need to wait until everything is
fully
observed but there's something that
explanation points
to that isn't just like an inventory of
what data were
collected otherwise would say it lets us
create an inventory of what we have
measured but an inventory isn't an
explanation so the form hypothesis is
actually in the interpretation of
perception absolutely absolutely
helmholtz perception as unconscious
inference it's like there's there's a
blur a red
blur and there's uncertainty but it's
like well if somebody is at the
racetrack maybe it's a red car maybe
they have a strong prior but that's a
hypothesis but maybe it turns out it was
actually just you know an interior
designer moving a Swatch of color in
front of their eye so then then later on
they they they up evaluate the
probability of that hypothesis how are
those hypotheses generated and winnowed
abductive
inference so whether we choose to kind
of um exalt it and make it clean and cut
and dry like a scientific
hypothesis or whether to blur it back
into the kind of imperceptible like
could a hypothesis does a hypothesis
need to be
experienced um if it doesn't and
one's willing to kind of relax that and

use hypothesis to include just um
proposed
causes proposed latent causes of
observations then that is exactly what
these accounts
are I mean what's what does the brain
do propose and select amongst hypotheses
again in this more broader sense not
just the scientist sense of
observations I'm my my stomach is
rumbling and this and that and this and
that maybe I'm in this physiological
state

Sharon oh go

ahead so one final question and uh and
it may come out in somebody else's
question but um but hidden

States

would would

invariably based on my understanding at
this point in time um lead to
probability so because if it's a hidden
State we can't know it and and um and so
there we have to kind of rely on what's
the

probability of this hidden State
impacting

the um the

hypothesis would that be

correct yes okay like just just to to um
to to make

hyperbole usually the data are taken
without

error and the hidden States always come
along with a probability distribution

this is just the simplest case like
thermometer says like 72 you might know
that it's like the most joke inaccurate

thermometer ever but you still take that
72 and you write it down your lab
notebook as
72 and then you have some
probability that's like okay well my
estimate of the temperature is actually
80 plus or minus 10 because I know this
thermometer is noisy and it low balls so
you have some probability distribution
for the hidden
state but then usually the data are
taken as simply the value that they
represent and that's reflected with the
Dirac Delta distribution that you'll
hear people talking about the Delta
distribution and that's basically a
special statistical distribution that's
like flat everywhere and then just has a
spike at the one Location
so that's kind of a way to coerce like a
data point into a distributional
framework that came up a lot in the
message passing live streams in 55 with
Magnus Umm Sharan or anyone else want to
like give a
thought
yeah hypothesis is used a lot of
times ranging
[Music]
from like
the view from the outside like talking
about the basing brain hypothesis as a
scientific
idea to this more like view from the
inside like the agent is hypothesizing
whether it is an apple or a frog in
their
hand here uncertainty about the hidden

state is being explicitly described
as a range of
so would the the Delta
distribution be a way to account for
that range of
hypothesis uh is that what is that what
you're
saying all go you get into so much
probability that you know then then
just like overwhelmed with probabilities
well yeah that can happen let's look to
um I'm going to the slides for for 55
because this is where like there's kind
of an interchange
between the
uh
okay check out the the papers or or 55
series for for more information but
basically what is happening on this is a
a a factor
graph so it it has a a onetoone
relationship with a basian graphs that
we look at in the
textbook um at time point one just
looking only at this one node that's
been kind of
highlighted um here there's an edge and
here there's a little Delta land like a
little token Delta that lands on that
edge so before um the data have been
collected at all this is like before the
first data point this is after the first
data point and this is after the nth
data point or the teth data point so
before this is kind of like an this is
like a a bungee chord that's not pinned
down there's just a full distribution
that relates these two

components then the data point comes in
and then that is seen as a
clamping so it's kind of like the bungee
cords waving around the probability
distribution could be within this range
and then the data point lands like the
dart lands and it just pins the cord
right
there and
then um to kind of keep the kind of that
thing the
same the Delta distribution is
used so it's just kind of like saying
we're going to clamp whatever
distribution we thought was going to be
there um instead of it being a gaussian
with 10 plus or minus one then the data
came in it was 11 now we have a derck
distribution with a parameter of 11 so
it kind of keeps that distributional
flavor and Keeps Us in this
probabilistic
setting and what's Happening Here is
basically before any data come in the
system is totally
unclamped and then as data points come
in those they clamp specific
parts of
graph and then inference proceeds on the
unclamped
Parts I don't know if it'd be helpful
but but think about like a guitar like
when you clamp down on a string and then
the rest of the string vibrates in a
different
way so an action has been taken or a
decision's been made I mean would be one
segment one example not necessarily the

only ones but in other words when she's
clamp down there's
movement yeah yeah actions have been
taken time by time step by time step
here specifically what's being
highlighted is like the sensory um data
point coming in but at each time step
it's not just that a sensory data point
comes in also that the action is
taken becomes irreversible to a certain
degree yeah like it um the past you
becomes
sedimented and reflects in the current
state of the
present because then you could say well
what if you forgot the past data point
well that's kind of like
unclamping the past and then now you
have another uncertainty distribution
over
memory like yeah what was the
temperature last
week I threw out the
data or could that represent you're
falling into a pattern or a habit
that's that's an an interesting question
also probably many ways to answer it but
like one way to look at it from
um from the kinds of models we looked at
in the textbook would be like um if we
were going
to if we were going to if we were gonna
have a a memory of um if we were gonna
have a memory of um we're we're P
pulling B balls out of an urn this is a
very common
statistical setting like you're pulling
a ball out of an

ear um and one one way to um you know
it's coming up you know orange orange
green orange orange green like
two-thirds of them are orange or
something so one kind of memory that you
might want to have would be
like remember which time step you pulled
which ball and then your memory register
is going to grow linearly accordingly to
how many polls that you make that you're
going to keep on recording the full data
so that's good for for to to that that
is the full data that's going to enable
the fullest analysis but let's just say
that you in a um computation limited
setting or maybe you wanted to respect
the privacy of previous polls then maybe
all that you would carry forward would
be like um 2third of all the polls that
I've made are orange and I've made a 100
poles so then you've coar grained over
like which specific polls were
which but you have now all you carry
forward is these two numbers how many
polls have you made and what fraction of
them were
Orange um so now you have a
smaller memory
register and that is exactly how one way
that people have modeled how habit is
learned so instead of the agent needing
to memorize the exact total sequence of
all previous
actions and then do this
reanalysis in the moment with like the
whole sequence of
action um you can
sediment previous actions into

habit and then you just carry the Habit
e forward as your policy
prior so that is one one lower
computational requirement
way to carry forward and slowly learn
habit thank you yeah that's interesting
though and then that
statistical setting is called the POA
earn there's so many variants of this
statistic
problem
but as as with many statistical problems
like we have
observations that's the color of the
ball and then we have our hidden State
we can't see inside the urn all we can
do is draw and so in this one um it's
not emphasizing the policy selection
planning
side like the the the the draw just is
given to you but um we could look at the
um the related
setting of a bandit or a multi-armed
bandit which is kind of like a
semi joking term for a slot machine I
guess because it steals your money
one-armed Bandit but the multi-armed
Bandit
is like a bunch of slot
machines and you don't know what's
inside of them slot machines like the
earn um now Which slot
machines should you use one strategy is
just sit down at the first one keep
using it another strategy is just rotate
through all of them equally another
strategy is um sit at them
proportionally to how rewarding they are

are another one is sit at them
proportionally to how confident you are
about your estimates in them or
inversely confident to how um inversely
timed to how confident you are in them
and so that slightly more active setting
of choosing where to draw observations
from that is where we've seen some very
useful active
for
um anyone who hasn't asked a question W
to it could be about any aspect of the
first half of the book or the whole
book
I have some project Associated questions
but I people with related questions
first
yeah let's look look at the
project here or
where well first um kind of off of a
question you asked during live stream
55 um where you were asking about active
inference
toolboxes
um do you have any thoughts or
recommendations if
if folks want to start playing around in
a computational setting on active
inference principles of what toolbox we
should start with or yeah good good
so few few answers first just just to
summarize in
55.1 or two or whatever it was like I
asked Magnus kud
all about the toolbox and about like the
ution of the toolbox and he gave a truly
classical highlights answer which is
like in terms of software

packages from the 90s to the early 2000s
it was all about mat lab language
statistical parametric mapping
toolkit with a special Focus from
neuroimaging created and maintained by
Carl friston's neuroimaging group at
UCL then then over the last 5
years as more machine learning connected
with active inference Connor Hines Alex
shance stafne dcus and others worked on
pmdp in
Python bringing many routines from SPM
into python setting and then most
recently with Bert of rise's group with
Magnus at all they are developing the
Julia toolkit which is even more General
than active inference you can do bayyan
settings that aren't active inference um
or are just super derived special cases
in terms of um where to start or look in
the code
page um
here there's like 30 or
40 different code
implementations for ACM and related I
always add whenever I find them and
anyone else can add them to
um if you are fluent in the language
then it may be enough otherwise even if
you're not familiar with a
language copying it into a language
model and asking for explanations might
be cool or asking to translate it into
pseudo code or give an example all these
other kinds of fun things so these are
um some specific places to
start
that's com kind of jumping in in the

computational end which is going to be perfect for

some I think the other toolbox that's now provided is chapter six and that's more like the um the um

Cafe sketchpad

toolbox like when you're sitting down to design an active inference model and it's not just well which computational architecture should we use but maybe we want to walk up to that point or

realistically develop an artifact if we address these questions I think we're already in a situation where this provides enough information to a language model to develop the computational generative model so we have the classical tool box elements and if somebody can just if somebody's already in the mechanic shop and can just get their hands dirty on these

tools they'll find more than enough and for those for whom just jumping into this kind of scripting isn't intuitive I think now we have a more

informal style that's almost conversational that

can get us there too especially on teams does that address it or what else would be

helpful that was helpful um follow on question do you know if versus um open sources or publishes any of their stuff

or I assume that's pretty proprietary
them off and I have not
like seen any specifically released code
but it may be but I'm I'm not
sure yeah it be added here as soon as I
find um in machine learning Street Talk
mentioned some something where they have
like an algorithm that can kind of
identify agents which is or
entities um that was very interesting to
me and something I was hoping to try to
make a um implementation of so that's
why I
asked yes interesting I remember that
too
Susan I'm in in their
um in versus
uh video which was kind of an
introduction they distinguish between
open Ai and um and their program as
being open
source so one thing you might want to do
Andrew is sign up for their
beta which is on their
website already
done yeah and so so we'll see how what
is open source and what's not yeah it is
an an interesting question what is open
source like just just not even with ACM
with language models like for some
people it's like you can use it it's
open source what's the issue whereas for
other people it's like no the Full
Source has to be provided the full data
set that was used for training and the
parameters that were used for training
and the weights of the model have to be
provided like Linux kernel the source

code is there and you can compile it
yourself that's what makes it open
source and then even further back
outside of the artifact of Interest like
is the mailing list Open Source can
people be observing or even included in
those discussions like those are all I
think very
interesting um I I'll add that um it if
um to understand how red hat makes money
might uh might inform you because you
what red hat U banks on is that what is
open source is not necessarily reliable
and so they've actually taken the open
source and turned it into something that
people can can depend
on so I don't if that's helpful or not
read where versus is going but I mean to
understand that model is
helpful yeah yeah definitely um
interesting and even even though we
expect and prefer the open source
materials to be reliable and anybody who
wants to for example have access to the
um textbook group repo at The Institute
GitHub that's
one great place if people just want to
play around with code and don't just
want it to be like a little abandoned
repo on their own but maybe want
to go in you know chapter by chapter
and cohort after cohort will iterate
towards having reliable pieces
like there will still be more and it's
not that it's our only um Guiding Light
but having reliable
open-source um accessories to the
textbook and to other educational

materials will be really
important um anyone who hasn't raised
anything
wna uh probably most of the ones with
get uh will be open source um I'm just
looking at uh RX infa and that's got a
um MIT
license oh yeah so uh the moment you get
on to
GitHub do you find that as good or bad
just out of
curiosity uh what the fact that it's it
or the or
MIT I've
Okay I uh my technical background as a
commercial coder I've never really
worried about the different subtleties
of MIT versus uh something that's
um uh
approved um is MIT less restrictive
about copying you can make closed Source
copies I
suspect um Okay so is not so maybe not
so good at supporting the open C open
source cuse but in terms of just being
able to use it I think that would be uh
I wouldn't have any worries about that
on my somewhat
limited um thank you
experience yeah interesting I also don't
know there's I know there's many flavors
of
Licensing um yeah this repo and their
documentation website is is
is extremely wellb built second in being
well-built would be
pmdp resources and
documentation and then kind of in a

class of its own would be the SPM
documentation which
has textbook and the
manuals and a list serve however those
discourses tend to be related
um human neuroimaging
experiments so you have less of like
people asking about
like how how do you do this like
theoretical agent identification or
something like that or some extension
and more of just like I'm using this
machine with this experimental setup how
do I get this to
work but yeah it's amazing how many are
out
here and so anyone who finds or makes
more please please add
them yeah pmdp has a BSD three Clause
license which again is um uh very open
source uh
so yeah here's the PDP
documentation and um follow the link to
GitHub you'll find uh the
license
cool yeah also this this documentation
this collab if people just want to run
they they don't even have a comp a
computer with a coding setup you can run
it in your browser and these are really
useful um
Andrew yeah I have a um unrelated
question that's um for you specifically
Daniel
um related to a paper you wrote in 2020
called active inference and behavior
engineering for teams
yep um has there been any Bon work this

is also very related to my
interests um so I was just wondering if
you have
a uh any you can
mention yes the follow on work was the
active inference
Institute
oh cool all right indeed then known
active inference lab basically just to
kind of close it this was September
2020 and then we thought okay where now
biggest issue was the accessibility
rigor and applicability of the framework
because getting to this artifact was not
so hard but to move
from multi-agent model model of of Team
narrative communication cyber physical
systems into having
them we were like
whoa how far even is
that hence the
lab
okay thank you
all it's been a great first
section will come back at the later time
next
week and then we haven't set the dates
or the times for second half of cohort 5
or for anything
else if people want to like explore any
role for a 2024 textbook group we can
talk about that so
just email if anything comes up
otherwise thank you and peace
bye and